

Cross-Patient Neural Speech Decoding via Atlas-Guided Spatial Calibration

Ben Tang | Cogan Lab, Duke University | April 2026

1 The Problem

Cross-patient speech decoding fails for a simple reason: patients do **not** share an electrode space.

Each patient has a 128- or 256-channel μ ECoG array placed by the surgeon on ventral sensorimotor cortex during intra-operative speech. Arrays differ in position, rotation, and coverage. Channel 17 in one patient is not the same brain location as channel 17 in another. Across patients, array centroids are offset by tens of millimeters, so a cross-patient model cannot rely on channel identity or grid position.

The problem is therefore: decode speech only after mapping different electrode layouts into the same functional space.

2 The Core Idea

The architecture separates cross-patient decoding into two problems:

1. **Spatial calibration** (per-patient): map each patient's electrodes into a common set of 16 cortical regions.
2. **Shared dynamics** (across patients): decode phoneme sequences from that common regional representation.

This gives a clean interface:

electrodes $\longrightarrow$ 16 regional activations $\longrightarrow$ phoneme sequence

The claim is not that patients share an electrode layout. The claim is that they share a **regional organization for speech**, and that the atlas gives a strong enough prior to calibrate each patient's partial observation into that common space.

3 Why This Should Work

Speech cortex has conserved large-scale organization. Face, tongue, sensory, Broca's, and auditory regions occupy a consistent order even though their exact coordinates vary across individuals.

Field potentials are geometry-driven mixtures. High-gamma at an electrode reflects nearby cortical activity weighted by distance, so the mapping is primarily geometric.

The atlas gives the right prior. Brainnetome provides anatomically meaningful speech-relevant regions rather than anonymous learned tokens. In a 10-patient regime, that prior matters.

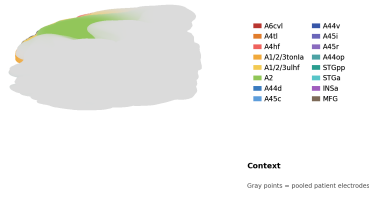
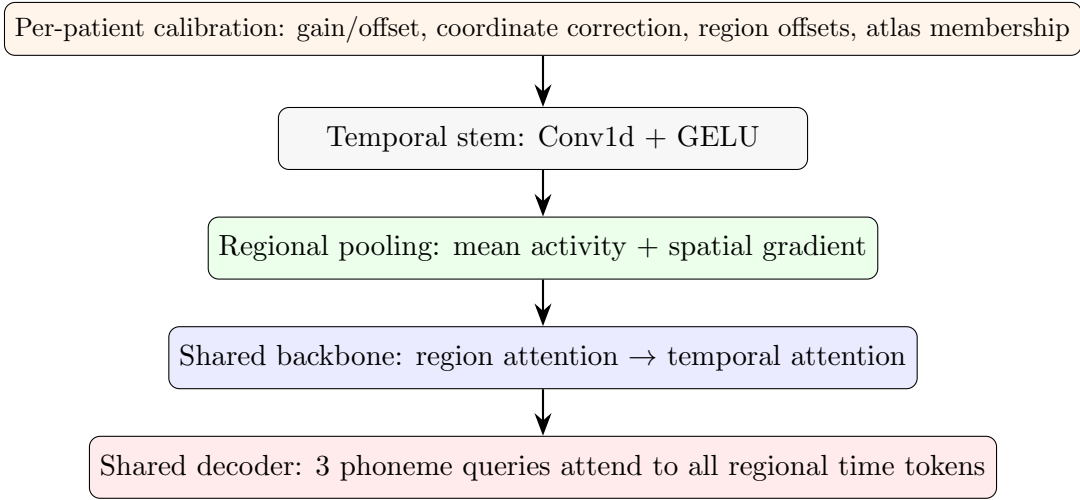


Figure 1: Brainnetome parcellation probabilities projected to the left fsaverage surface. Colors mark the 16 active speech-relevant regions; faint gray points show pooled patient electrode coverage.

4 Model in One Picture



Every patient first gets a small calibration layer that says where their electrodes sit relative to the atlas. After that, the model is shared.

5 Problem 1: Spatial Calibration

Spatial calibration is a **template-fitting problem**. The atlas provides the template; patient-specific parameters make bounded corrections.

Signal calibration. Each channel gets a learned gain and offset so the shared model sees comparable amplitudes across patients.

Global coordinate correction. A small rigid correction absorbs systematic error from surgical reconstruction and the linear MNI transform.

Region adaptation. Each of the 16 atlas regions gets a small patient-specific offset and a scalar temperature.

Volumetric membership. Electrodes are mapped to regions using Brainnetome parcellation volumes, not spherical kernels.

After calibration, every patient is represented in the same 16-region frame.

6 Problem 2: Shared Dynamics

Once the data is in a common regional space, the rest of the problem is to learn the shared dynamics that map regional activity to phoneme sequence.

Temporal stem. A small Conv1d converts each calibrated electrode trace into a feature sequence.

Regional pooling. Electrode features are compressed into 16 regional signals using average activity plus a first spatial gradient.

Shared backbone. Attention first models interactions across regions, then dynamics across time. This is where cross-patient sharing happens.

Shared decoder. Three learned queries read out the 3 phoneme positions by attending jointly across all regions and time steps.

7 What Is Learned Per Patient vs Shared

Component	Shared?	Role
Gain / offset, coordinate correction, region offsets, region temperatures	No	Calibrate where this patient's signals land in the atlas space
Temporal stem, regional pooling weights, spatiotemporal backbone, decoder	Yes	Learn what speech dynamics look like once signals are aligned
Per-patient parameters	326 (128-ch)	Small calibration layer
Shared parameters	~143K	One common speech decoder across patients

Per-patient parameters learn **where** speech information lives in this patient. Shared parameters learn **what** that aligned activity means.

8 Key Risks / Decisive Tests

Nonlinear MNI normalization. The current coordinate pipeline is still linear.

Atlas quality. We are using smoothed atlas maps as the working representation.

Decisive tests. The architecture stands or falls on a few comparisons:

- atlas regions vs learned bottleneck
- parcellation pooling vs cross-attention routing
- fixed atlas vs patient-specific region offsets
- pooled readout vs joint cross-attention decoder

9 Immediate Next Steps

1. **Implement the first cross-patient training loop.**
2. **Improve spatial accuracy.** If T1 MRIs are available, nonlinear MNI normalization should reduce the burden on the calibration layer.
3. **Scale shared dynamics with more data.** External chronic ECoG is most useful for pretraining the shared dynamics module, not for replacing atlas-guided calibration.